

High-speed compact silicon photonic Michelson interferometric modulator

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Abstract: We present the detailed analysis and characterization of a silicon Michelson modulator with short 500 μm phase shifters and a low $V_\pi L_\pi$ of 0.72 V-cm under reverse bias. We investigate optical modulation of reverse biased p-n and forward biased p-i-n junctions. We demonstrate for the first time that error-free operation up to 40 Gbps is possible with lumped silicon interferometric modulators. For reverse bias operation, we show that even greater bandwidth can be obtained with lower impedance drivers. Forward bias operation with pre-emphasized signals is shown to have clean eye diagrams up to 40 Gbps, however, error counting reveals a strong dependence on test patterns and that error-free operation is achievable for short pattern lengths.

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OCIS codes: (130.3120) Integrated optics devices; (250.4110) Modulators; (200.4650) Optical interconnects.

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1. Introduction

Substantial increase in the supply and demand of cloud based services and internet access media over the last decade has lead to a significant growth in bandwidth requirements. This trend will continue in the future, and with warehouse-scale datacenter networks, intradatacenter communication links are becoming longer [1]. For such communication links, optical interconnects have potential benefits with respect to density and energy compared to electrical interconnects [2]. These same issues are also present in high performance computing systems, and optical interconnects based on silicon photonics are considered a viable solution [3]. Hence, tremendous research is being carried out to optimize and innovate silicon photonic modulators using CMOS compatible processes to develop low-cost, energy efficient, and high bandwidth short reach interconnects.

In silicon, the electro-optic effects (Kerr, Pockels, and Franz-Keldysh) are relatively weak, therefore changes in the real and imaginary parts of the refractive index must be obtained by the plasma dispersion effect [4]. Typically, carrier concentration is modulated using a diode-waveguide structure in which carriers are injected (forward bias operation) or depleted (reverse bias operation). Reverse bias operation is preferred to forward bias operation because of reduced losses from lower optical scattering and negligible diffusion capacitance resulting in a larger intrinsic bandwidth.

Resonance based modulators such as micro-rings and micro-disks modulators have gained immense interests due to smaller requirements in index change leading to smaller footprints, and better modulation energy efficiency compared to broadband modulators. However, these modulators need to be tuned to a specific wavelength, and this is typically achieved with the thermo-optic effect along with additional electronic feedback control of the heaters [5, 6].

Alternatively, interferometric modulators such as the Mach-Zehnder modulator (MZM) are being investigated for short reach optical interconnects. MZMs are optically broadband, and

limited only by the bandwidth of the coupling devices. Because their modulation is relatively weak, long phase shifters are required which leads to a voltage-length trade-off for such modulators. A longer modulator will require a traveling wave electrode design and operate at lower drive voltages for a given optical modulation amplitude (OMA). Shorter modulators require larger voltages for a similar OMA but can be effectively studied using lumped element circuit analysis.

In Michelson interferometric modulators (MIMs) light travels through the phase shifting region twice compared to MZMs because of loop mirrors placed at one end of the modulator. When compared to nearly identical MZMs, MIMs have been shown to halve the $V_\pi L_\pi$ when operated at DC voltages [7]. This is most effective for lumped devices because the efficiency benefit deteriorates in traveling wave designs where there are counter-propagating RF and optical waves [8]. To date, very little work focusing on lumped modulators built in silicon and operating in reverse bias has been reported, and demonstrated designs operate at bitrates of approximately 10 Gbps [9, 10]. Higher rates under forward bias operation have also been demonstrated, but with the use of pre-emphasized signaling [11, 12].

We investigate compact interferometric modulators as they allow for denser integration and stable operation with heat dissipating electronics. In this paper, the operation of a relatively low $V_\pi L_\pi$ dense MIM fabricated on silicon-on-insulator (SOI) is demonstrated. We report on the DC, small signal, and large signal characterization of the modulator under reverse and forward bias with p-n and p-i-n junctions. We verify that the reverse bias modulation bandwidth is primarily limited by the 50 Ω source impedance. Open eye diagrams up to 40 Gbps for both forward bias with pre-emphasized signals and reverse bias are also shown. In addition, error free operation ($BER < 10^{-12}$) of the lumped modulator at bitrates up to 40 Gbps is demonstrated.

2. Design

A schematic of the fabricated MIM is shown in Fig. 1(a). The modulator uses an adiabatic 3-dB directional coupler for the input and output (I/O) ports [13]. The loop mirrors are formed with a compact Y-branch and bent channel waveguides. The compact Y-branch cell was reproduced from the parameters provided in [14]. Vertical surface coupling, with focusing grating coupler designs from [15], is used to couple light in and out of the chip.

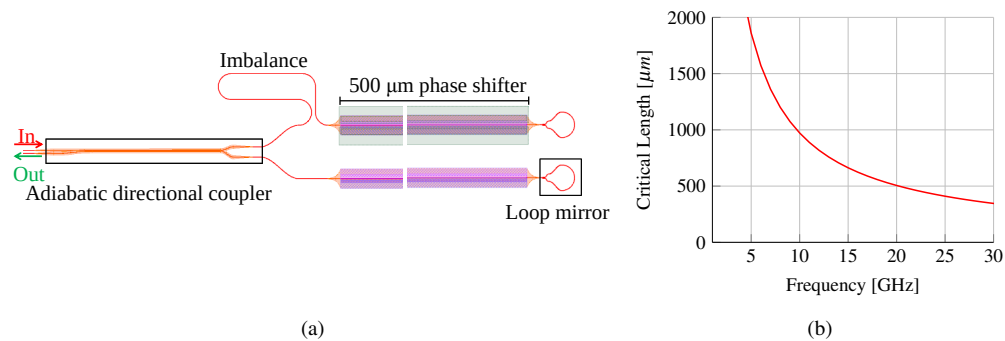


Fig. 1. (a) Schematic of the Michelson interferometer modulator, and (b) simulated critical length for lumped element analysis.

The modulator was designed to operate in a single drive configuration, thus metalization was performed on only one arm of the interferometer. Both arms of the interferometer were doped to equalize losses and optimize the extinction ratio (ER). An imbalance of 150 μm was included to help with phase shift measurements. Due to this imbalance and the lack of additional thermal

phase shifters, we operate the modulator near, but not exactly at, 1550 nm.

The phase shifter length was chosen to be smaller than 1/10th of the effective microwave wavelength λ_{eff} for lumped element modeling and analysis. Figure 1(b) shows the critical length for this rule of thumb, $L_{critical} = \lambda_{eff}/10$, as a function of frequency. The microwave effective wavelength for a quasi-TEM mode with coplanar strip electrodes loaded with a p-n junction was obtained from simulations in ANSYS HFSS. This simulation considered the frequency dependence of the effective index, which at 20 GHz was found to be 3.0 using estimated resistance and capacitance of the p-n junction at 0 V and 2.1 for the case of the unloaded electrode. The phase shifter was designed to be 500- μm long in order to be considered electrically small up to at least 20 GHz. In addition, it is assumed that the optical transit time is much shorter than the microwave travel time.

MIM designs were fabricated using a commercially available process at the Institute of Microelectronics (IME) A*STAR (Singapore) on a SOI wafer with a 2 μm buried oxide (BOX) layer and a 220-nm thick top-silicon. This process had a single 2- μm thick aluminum interconnect metal layer, aluminum vias, and non-silicide contacts. The designs used 130 nm etch steps for creating rib waveguides and adiabatic directional couplers, 220 nm deep etch for channel waveguides, and 70 nm etch steps for focused grating couplers.

Modulating diodes were developed in both p-n and p-i-n junction configurations. Figure 2(a) shows the p-n junction geometry and Fig. 2(b) shows the p-i-n junction cross-section. Higher concentration dopants (p++ and n++) are used for ohmic contacts and the lower concentration dopants (p and n) are used in the region of the waveguide where the optical field is most concentrated. Similar to [7, 9], an offset of 100 nm relative to the center of the waveguide was used in the p-n junctions to increase the mode overlap with the p-type semiconductor. This offset was expected to improve the modulation efficiency because the plasma dispersion effect is more sensitive to changes in free hole concentration [4].

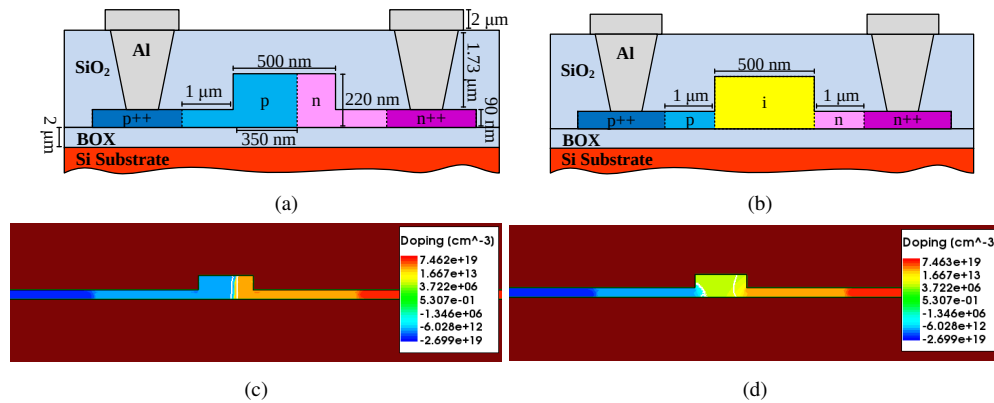


Fig. 2. Cross-section of the fabricated (a) p-n, and (b) p-i-n MIM. Simulated depletion region (delineated by white lines), and doping density (positive for n-type) of the (c) p-n and (d) p-i-n junctions at 0 V.

The dopant concentration and distribution, specific to this process, was simulated using the foundry implant and annealing conditions with the Synopsys Sentaurus Process software. The process simulation used Synopsys's advanced calibration parameters. The simulations showed peak p++, p, n, and n++ dopant active concentrations of 3.9×10^{19} , 7.8×10^{17} , 2.1×10^{18} , and $9.7 \times 10^{19} \text{ cm}^{-3}$, respectively. The dopant distribution and concentration for the p-n and p-i-n junction is displayed in Figs. 2(c) and 2(d), respectively. A device simulation was subsequently performed using Synopsys Sentaurus Device to obtain the distribution of free carriers in the

waveguide for different bias voltages. This tool solved the drift-diffusion equations for electron and hole densities coupled with the Poisson equation for the electrostatic potential, and Fermi statistics. Other activated physical models in this simulation were incomplete ionization, Shockley-Read-Hall recombination, and doping-dependent mobilities. The change in bulk index was then calculated by applying Soref and Bennett's fitted model to this simulated distribution. The bulk index profile was then imported, for each voltage, in Lumerical MODE to calculate the optical effective index. A similar simulation methodology has been recently reported in [16]. Our simulation results, however, showed marginal improvements in efficiency with the 100 nm offset p-n junction in this process. As shown later in section 3.1, the phase shift simulated using this approach matched reasonably well to the measured phase shift of the device we tested, indicating that this simulation flow can be used to estimate the behavior of future modulator junction designs.

3. Experiment and results

3.1. DC characterization

The current-voltage (I-V) relationship of the diodes were measured. The stimulus was applied to the modulator with a ground-signal-ground (GSG) probe. The reverse bias and forward bias I-V curves are shown respectively in Figs. 3(a) and 3(b) for both the p-n and p-i-n junctions. The forward bias curves indicate that both types of diodes begin to turn on at about 1 V. Under reverse bias operation, it was observed that the p-n junction reaches breakdown at approximately 8.5 V. This breakdown voltage was observed to increase with temperature, thus affirming avalanche breakdown as the mechanism. The p-i-n diode, however, does not breakdown in the measured voltage range. The occurrence of the breakdown imposes limitations on the bias and driving voltages because a higher instantaneous.

The optical transmission spectra, shown in Figs. 3(c)-3(f), were measured for the p-n and p-i-n junction under different bias conditions. Light was coupled to the chip using a fiber array. The fiber-to-fiber I/O coupling loss was measured to be 8.8 dB. The routing loss from the grating couplers to the modulator I/O was estimated to be 1.2 dB. Removing these contributions from the transmission spectra measurements, the insertion loss for the modulator with the p-n junction is determined to be approximately 4.7 dB. An optical extinction ratio of 30 dB was measured at 0 V indicating a good loss balance between the two arms of the interferometer. The p-i-n junction modulator was measured to have a lower insertion loss because of the wider depletion region at the center of the waveguide as illustrated in Fig. 2(d). Its loss was estimated to be 3.7 dB. There is a loss of approximately 3 dB intrinsic to the MIM because modulated light on the return path is split at the directional coupler and sent towards the source. In our experiment, an external isolator was not required because there was one present inside the laser.

The phase shift of the p-n junction modulator under reverse bias was calculated using $\Delta\phi = 2\pi\Delta\lambda / FSR$, where $\Delta\lambda$ is the wavelength shift of the spectrum relative to the 0 V reference, and the FSR is obtained from the measured transmission spectrum of 0 V, as depicted in Fig. 3(c). The phase shift is plotted in Fig. 4(a), and it is observed that the amount of phase shift is non-linear and decreases with increasing voltage. This occurs because the successive changes in the depletion region width overlap less with the optical mode, and also because the depletion width varies as the square root of applied voltage. Similar to [17], the $V_\pi L_\pi$ was calculated rather than measured from devices with different phase shifter lengths. The values were obtained from $V_\pi L_\pi = \pi V_{applied} L_{DUT} / \Delta\phi$, where $V_{applied}$ is the applied reverse bias voltage, $\Delta\phi$ is the deduced phase shift at that voltage, and L_{DUT} is the phase shifter length (500 μm). The calculated $V_\pi L_\pi$ is plotted in Fig. 4(b). Both figures illustrate the diminishing change in phase for larger voltages, and that there is a trade-off between modulation efficiency and operation at higher bias voltages

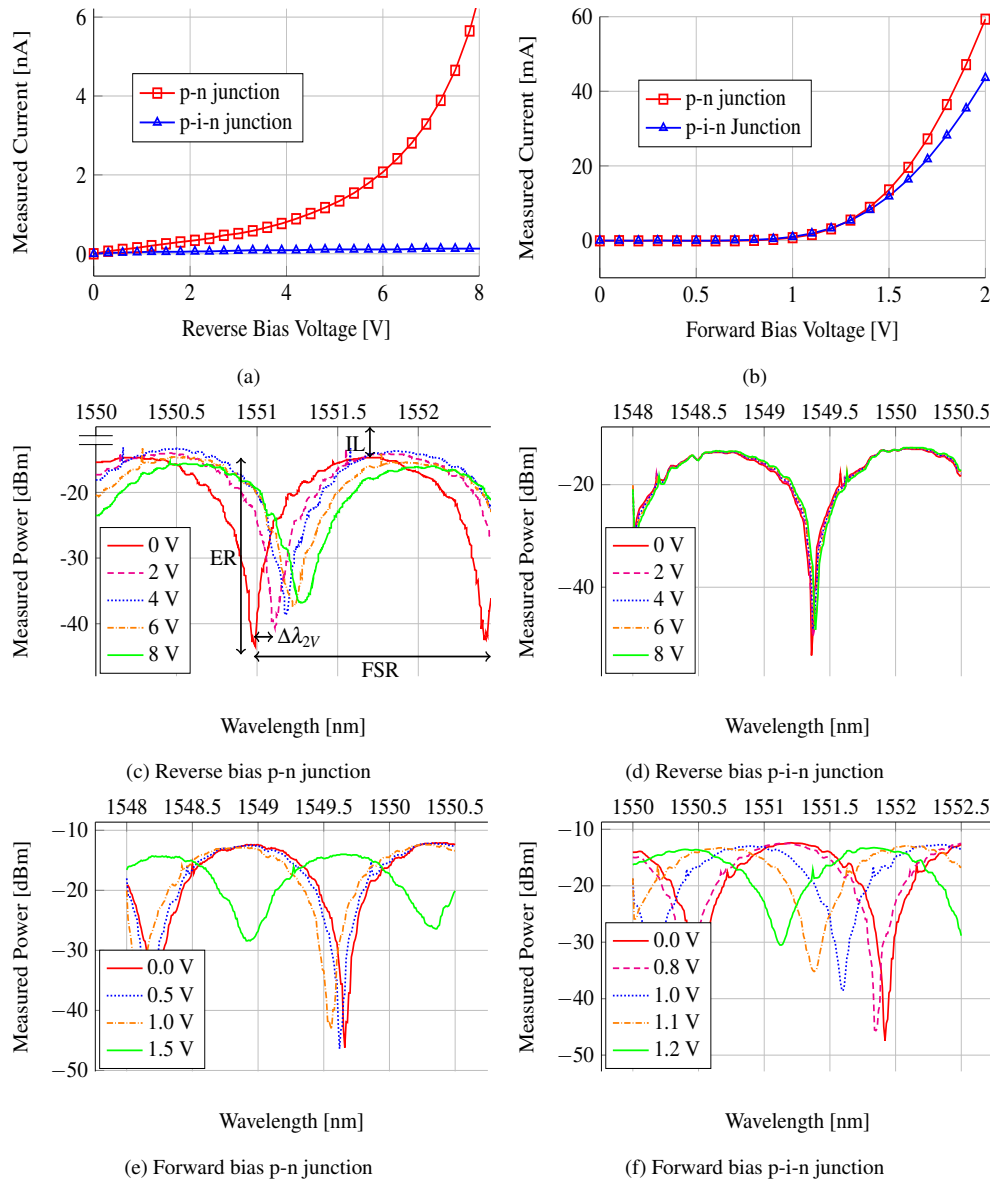


Fig. 3. I-V measurement of p-n and p-i-n junctions with (a) reverse bias, and (b) forward bias. Measured transmission spectrum under reverse bias of (c) p-n and (d) p-i-n MIM. Measured transmission spectrum under forward bias of (e) p-n and (f) p-i-n MIM.

for greater bandwidth. We also observed a good agreement between the measured result for this device and data obtained from simulations using the tools and methods described in section 2. It is important to note that because of the non-linear nature of the phase shift as a function of voltage, the $V_\pi L_\pi$ figure-of-merit does not allow the accurate calculation of V_π for a given modulator length or L_π for a given voltage.

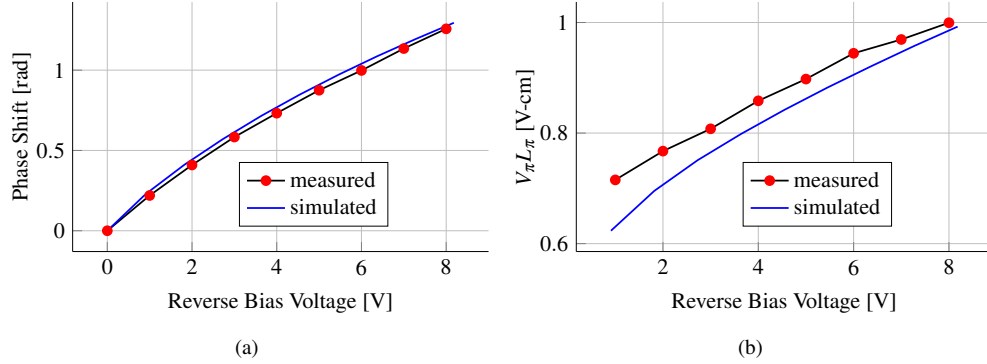


Fig. 4. Derived from measurement and simulated (a) phase shift, and (b) $V_\pi L_\pi$ of the p-n MIM with reverse bias voltage.

The p-i-n junction based modulator under reverse bias shows negligible change in phase because of the large depletion region already overlapping the optical mode at equilibrium conditions. The optical mode sees minute differences from any additional carrier extraction with applied voltage. In forward bias, insignificant phase shifts are observed below threshold. A π phase shift is observed with relatively small voltages after threshold because of the large number of injected carriers. The large carrier injection also results in a greater loss, and for this single drive configuration produces an imbalance in losses between the two arms of the MIM. This leads to a reduced extinction ratio in the interferometer as represented by the transmission spectra. From Figs. 3(e) and 3(f), it is observed that the p-i-n junction device reaches π phase shift at a smaller forward bias voltage compared to the p-n junction based modulator. The p-n junction modulator has a π phase shift at 1.5 V forward bias, corresponding to a $V_\pi L_\pi$ of 0.075 V-cm, whereas the p-i-n junction modulator has a π phase shift at 1.15 V corresponding to a $V_\pi L_\pi$ of 0.058 V-cm. Besides optical modulation, forward biased MIM can be of interest in switching applications, for instance, it can substitute the thermo-optic based MI switches demonstrated in [18].

3.2. Small signal characterization

The RF S_{11} parameters were measured using an Agilent 50 GHz lightwave component analyzer (LCA). The RF power on the LCA was set to -5.0 dBm corresponding to $0.356 V_{pp}$ in a 50Ω load, which is small enough to approximate the non-linear electro-optic (EO) response to a linear one. In addition, the wavelength was set to bias the modulator at quadrature point to ensure linearity, and reduce spurious peaks and dips during measurement. The measured RF S_{11} and EO S_{21} for different reverse bias voltages is shown in Fig. 5. A 3-dB bandwidth of 13.3 GHz is measured at 4 V bias. Marginal improvement in the 3-dB EO bandwidth is observed for reverse bias greater than 4 V because of smaller reductions in depletion capacitance for higher biases.

The S_{11} curves show high reflection, particularly at lower frequencies, which occurs because the p-n junction under reverse bias is essentially a capacitor that behaves as an open circuit at low frequencies. Since it is the voltage across the capacitance which performs the optical

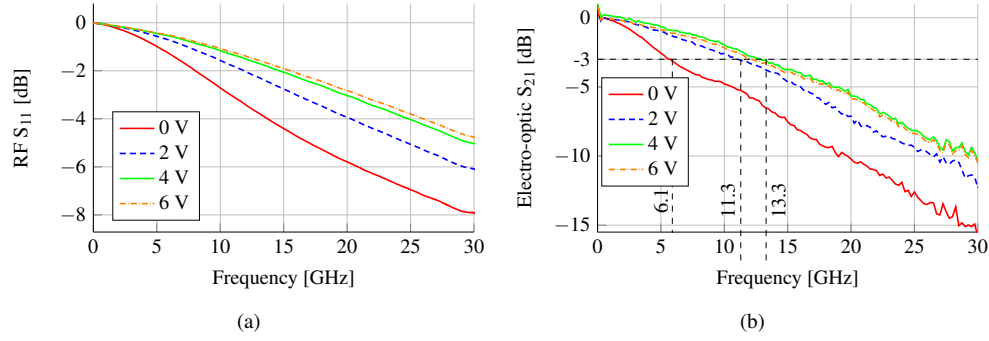


Fig. 5. Magnitude of measured (a) RF S_{11} , and (b) EO S_{21} for different reverse bias voltages.

modulation, high reflection is expected and not indicative of poor modulation. In traveling wave designs, low return loss is observed because power is transferred to and consumed by the matched termination. Additionally, reflections are reduced at interfaces with the matching of the transmission line (i.e., the modulator itself) with the source and termination.

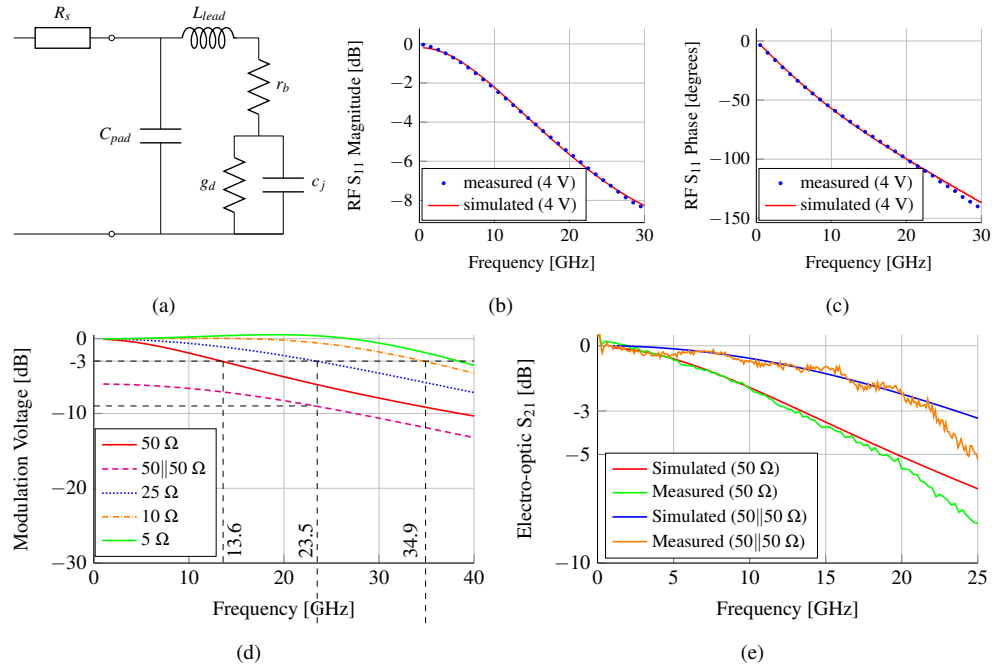


Fig. 6. (a) Small signal equivalent circuit of a lumped reverse biased modulator. Simulated and measured (b) magnitude, and (c) phase RF S_{11} of the p-n junction with 4 V reverse bias. (d) Simulated EO S_{21} response at 4 V reverse bias for different source impedances. (e) Comparison of simulated and measured EO S_{21} responses for two cases at 4 V reverse bias.

In order to understand the bandwidth limitation of the modulator in this process and to be able to estimate the energy consumption in section 3.6, the values of the circuit elements in a lumped element model had to be determined. A small signal model of the modulator in reverse

bias is shown in Fig. 6(a), where C_{pad} is the parasitic capacitance between the signal and ground pads, and L_{lead} is the parasitic lead inductance [17]. The other circuit elements represent a small signal model of a reverse biased diode [19], where r_b is the contact and quasi-neutral region resistance, g_d is the conductance of the depletion region, and c_j is the depletion capacitance. Additionally, the source resistance R_s is also shown in Fig. 6(a). The substrate parasitic capacitance and resistance is ignored in this model because the relatively thick BOX layer, illustrated in Fig. 2, leads to a small capacitance, and the high resistivity handle substrate leads to a high resistance. The parameters of the small signal model were determined empirically using the S_{11} measurements. As Figs. 6(b) and 6(c) show, there is a good agreement between the measured and simulated S_{11} curves. This confirms that the modulator can be analyzed with lumped components in the measured range of frequencies.

The extracted small signal parameters for the p-n and p-i-n junction are listed in Table 1. As expected, c_j decreases with increasing reverse bias because of a larger depletion width. The larger depletion width also reduces the transverse length of the quasi-neutral region, thus lowering r_b with higher reverse bias voltages. For a p-i-n junction, the much larger depletion width results in a much smaller c_j and r_b leading to a large modeled junction bandwidth. The rise time in Table 1 was calculated considering the semiconductor junction alone, and modeled as a series r_b and c_j circuit only. Then, for this single-pole network, the 10-90% rise time is $t_r \approx 2.197r_b c_j$.

The extracted parameters were validated by performing an AC simulation using a 50 Ω source impedance and the model in Fig. 6(a) with the parameters of Table 1. The simulated 3-dB bandwidth of the voltage across the depletion capacitor c_j at 4 V reverse bias with the 50 Ω source impedance is shown in Fig. 6(d). The simulation matched the measured EO S_{21} 3-dB bandwidth of Fig. 5(b) within 0.3 GHz. For comparison, the simulated responses with a source impedance of 50 Ω alone and then with a 50 Ω parallel resistor, and their corresponding measured EO S_{21} responses from Fig. 5(b) are re-plotted in Fig. 6(e). Although the modeling parameters are unique to this modulator design, they provide insight into the capabilities of this process. Furthermore, the good agreement between simulation and measurement indicates that this modeling approach can be applied to future lumped modulator designs.

Table 1. Small signal circuit model for reverse biased junctions.

	C_{pad} [fF]	L_{lead} [pH]	r_b [Ω]	c_j [fF]	$1/g_d$ [k Ω]	Junction BW [GHz]	t_r [ps]
p-n (0 V)	14.0	139.3	32.9	259.7	26.4	18.6	18.8
p-n (-2 V)	13.0	141.8	29.6	189.0	15.7	28.4	12.3
p-n (-4 V)	12.2	140.5	28.1	168.9	11.7	33.5	10.4
p-n (-6 V)	13.2	138.5	28.6	158.2	9.2	35.1	9.9

3.3. Large signal characterization

The large signal behaviour of the modulator was studied as part of a communication link illustrated in Fig. 7. A SHF bit pattern generator (BPG) was used to obtain a 0.4 V_{pp} PRBS-31 signal. The signal was amplified with a 42 GHz SHF RF amplifier. In order to prevent the diode from entering the breakdown regime and to limit the driving voltage, a 6 dB attenuator was placed at the output of the amplifier. The resulting driving voltage was measured to be 3.4 V_{pp} at 10 Gbps dropping down to 3 V_{pp} at 40 Gbps in a 50 Ω load. A 65 GHz bias tee was used to offset the PRBS signal to a reverse bias of 4.5 V. This signal was then applied to the modulator by using a GSG probe.

For the optical link, a tunable laser source was used to provide light at 1550 nm. Light was coupled to the silicon photonic chip (SiP) using a fiber array, and vertical focusing grating

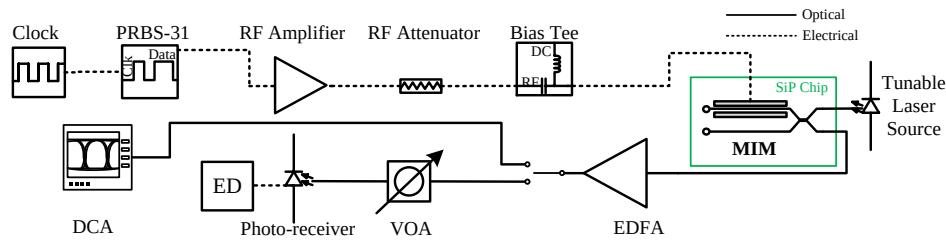


Fig. 7. Block diagram of a testing system for a short-reach communication link with a silicon photonics modulator.

couplers. The modulated optical signal was amplified with an EDFA. For sensitivity curves, the modulated optical signal was attenuated with a variable optical attenuator (VOA), captured by a 35 GHz Picometrix photo-receiver, and connected to an error detector (ED). Optical eye diagrams were collected by temporarily connecting the modulated light to an Agilent digital communications analyzer (DCA).

The optical eye diagrams at different bitrates with a PRBS-31 signal are shown in Fig. 8. The ER and the Q-factor, as measured by using the DCA with the optical sampling module, are listed in the sub-caption. In a typical optical communication system, a Q-factor of 7 corresponds to a BER of 1×10^{-12} . The Q-factor reported by the DCA, however, is much more pessimistic due to the noise of the sampling module. Therefore, the Q-factor reported here is solely for qualitative comparison between the different modulation cases presented in the paper. Clear eyes were observed up to 30 Gbps. At 34 Gbps, significant distortion is observed in the eye diagram. In Fig. 9(a), the optical extinction ratio reported by the DCA for different bitrates is plotted. The wavelength of the laser was set to operate the modulator at quadrature point for these measurements.

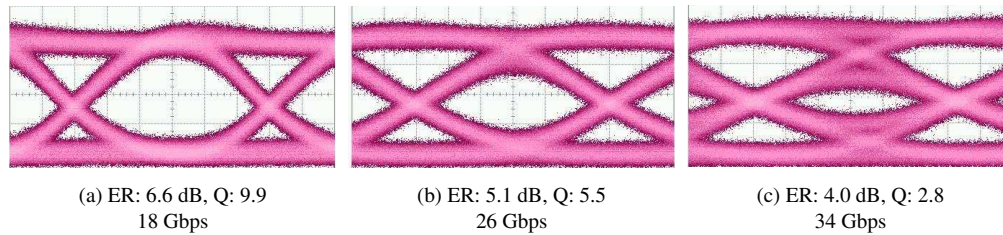


Fig. 8. Optical eye diagrams at different bitrates.

The sensitivity curves for the MIM with the p-n junction is shown in Fig. 9(b). The 0 errors line indicate a measurement with 0 errors detected in 3 terabits, thus allowing us to claim error-free operation ($BER < 10^{-12}$) with a 95% confidence level [20]. The link operates error-free at 25 Gbps with a received power of -5 dBm and at 30 Gbps with a received power of 1 dBm. A larger driving voltage would improve the sensitivity, illustrating a trade-off with power consumption.

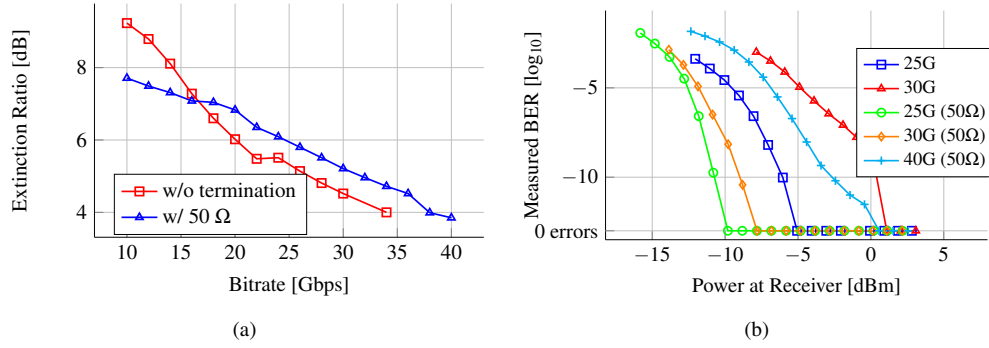


Fig. 9. (a) ER measurement using the DCA, and (b) sensitivity curves for operation with and without a 50 Ω parallel resistor at different bitrates.

3.4. Bandwidth extension with a parallel resistor

In a lumped model, the effective optical modulation is given by the voltage drop across the junction capacitor c_j . The 3-dB cut-off frequency, ignoring the parasitic elements, is simply $f_{3dB} = 1/[2\pi c_j(r_b + R_s)]$, where R_s is the impedance of the source seen through the coaxial transmission line, typically 50 Ω. From the extracted parameters of Table 1, it is observed that R_s is typically larger than r_b , thus the bandwidth of a lumped modulator can be improved by placing a resistor in parallel as shown in Fig. 10(a) to reduce the equivalent source resistance and obtain: $f_{3dB} = 1/[2\pi c_j(r_b + R_t||R_s)]$. This was experimentally verified by using an SGS probe and placing a 50 Ω RF termination on one of the SG pairs. The measured EO S_{21} for different reverse bias voltages is shown in Fig. 10(b). At 4 V bias, an improvement of 10 GHz is observed compared to the case measured in Fig. 5. This 23 GHz bandwidth is theoretically sufficient for transmitting the fundamental component, attenuated by 3 dB, of an alternating NRZ bit sequence at 46 Gbps [21].

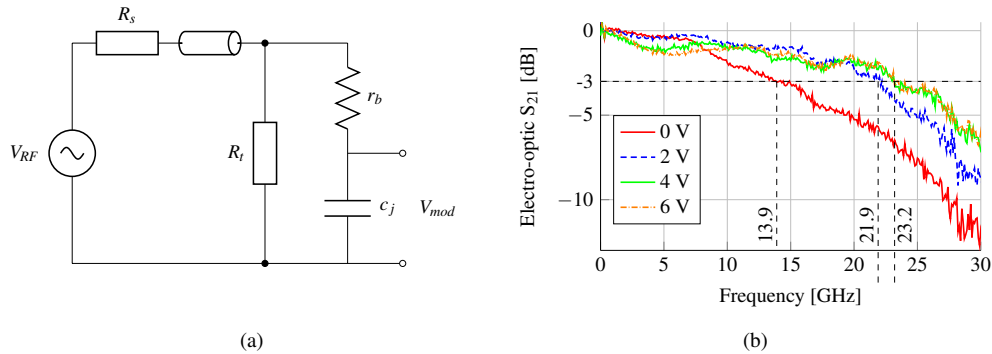


Fig. 10. (a) Equivalent circuit model with a parallel resistor, and (b) magnitude of measured EO S_{21} for different reverse bias voltages with a 50 Ω parallel resistor

The optical eye diagrams for different bitrates with this driving configuration are shown in Fig. 11. The ER and the Q-factor, as measured by using the DCA with the optical sampling module, are listed in the sub-caption. Clear eyes were observed up to 40 Gbps indicating a significant improvement in performance compared to the eyes shown in Fig. 8. In Fig. 9(a), the optical extinction ratio, as measured with the DCA, for different bitrates is shown. The driving signal was identical to the one used in the case without the parallel resistor. However,

the presence of the parallel resistor reduces the voltage drop across the depletion capacitance. It is observed that the ER at low bitrates is smaller than the case without the parallel resistor. The ER is, however, slightly larger for higher bitrates. This happens because at higher bitrates the RF signal is more attenuated for the case without the parallel resistor due to the limited bandwidth.

The sensitivity curves for the MIM with the p-n junction and a $50\ \Omega$ resistor in parallel is shown in Fig. 9(b). The link operates error-free at 25 Gbps with a received power of -9.8 dBm, at 30 Gbps with a received power of -7.8 dBm, and at 40 Gbps with a received power of 0.6 dBm.

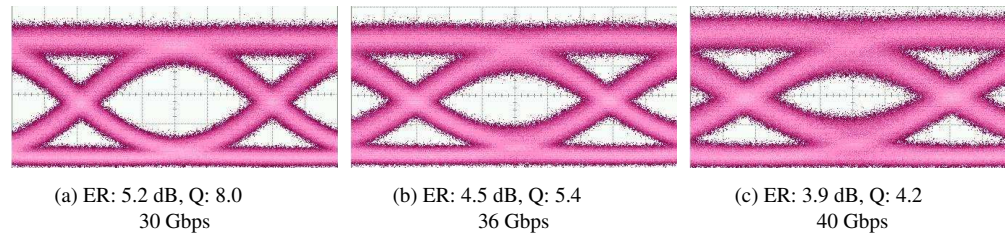


Fig. 11. Optical eye diagrams at different bitrates for the modulator driven with a $50\ \Omega$ parallel resistor.

An AC simulation using a $50\ \Omega$ source impedance, a $50\ \Omega$ parallel resistor, and the model in Fig. 6(a) with the parameters of Table 1 was performed. For this case as well, the simulated 3-dB bandwidth of the modulation voltage shown in Fig. 6(d) of 23.5 GHz matched the measured EO S_{21} bandwidth in Fig. 10(b) of 23.2 GHz at -4 V bias. Re-simulating with a $25\ \Omega$ source impedance without a parallel resistor showed the same EO S_{21} bandwidth but with a larger voltage drop across the depletion capacitance leading to a better overall modulation efficiency. Re-simulating with a $10\ \Omega$ source impedance showed a EO S_{21} bandwidth of 34 GHz without any peaking in the response. In this case, the simulated bandwidth, which also considers the parasitic inductance and capacitance, corresponds to the extracted junction bandwidth. With a further reduction in source impedance, even a larger bandwidth can be observed but with peaking. These simulations and experimental results suggest that the bandwidth limitation of lumped modulator is mainly due to the source impedance. It is noteworthy that higher bitrates can be attainable with appropriately designed drivers and transmission lines. Furthermore, higher speeds without the additional parallel resistor will reduce energy consumption.

3.5. Forward bias with pre-emphasis

High bitrate operation in a poor channel is typically achieved using digital-signal-processing to perform pre-emphasis with a digital-to-analog (DAC) converter and/or equalization with an analog-to-digital converter. Using this approach, we recently demonstrated the first traveling wave silicon photonic intensity modulator operating at 112 Gbps transmission over 10 km of SMF fiber operating below pre-FEC BER threshold of 3.8×10^{-3} employing PAM-8 at 37.4 Gbaud [22, 23]. By putting emphasis on transitions, high-frequency content is given a larger magnitude than the low-frequency content. A simpler low-cost method to using a DAC involves creating a pre-emphasis signal using analog components that is equivalent to a feed forward equalizer (FFE) with a small number of taps [12]. To demonstrate the high-speed operation of a forward bias p-i-n MIM, a two-tap FFE was created as depicted in Fig. 12(a). The synchronized data from one channel and the inverted data from another channel of the BPG was power combined, amplified, and then attenuated. The amplitude of the data signals and the delay between them were controlled with the BPG.

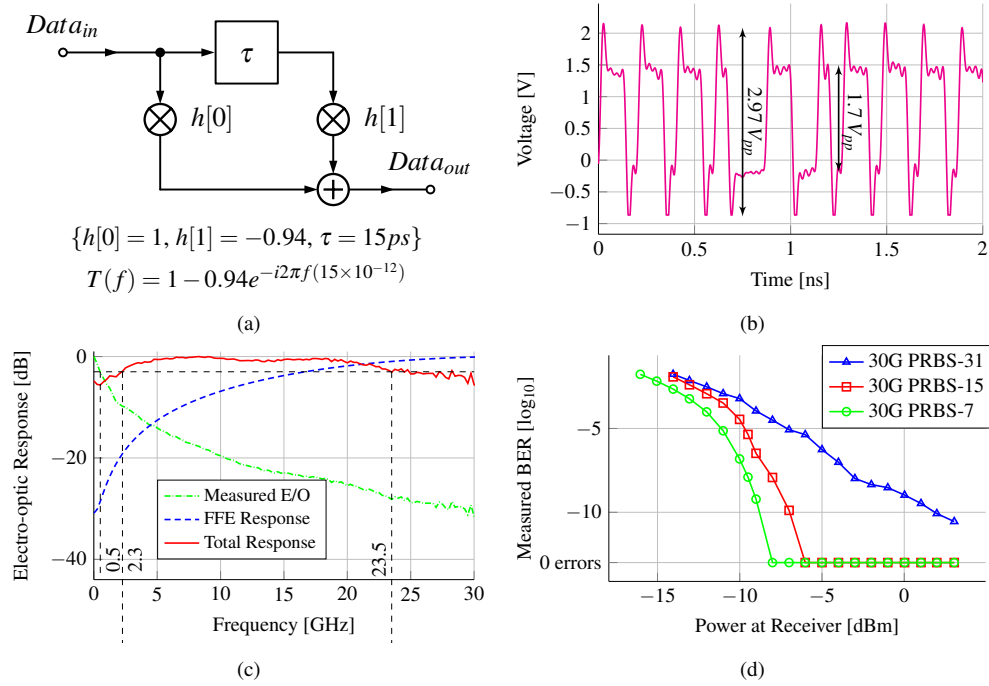


Fig. 12. (a) Two-tap feed forward equalizer for pre-emphasis of data signals. (b) Pre-emphasized driving signal. (c) Measured EO response at forward bias, and calculated FFE and total responses. (d) Sensitivity measurement for different pattern lengths for forward bias operation.

The resulting emphasized driving signal, shown in Fig. 12(b), had a voltage of $2.97 V_{pp}$, and the diode was forward biased to 0.6 V. The voltage without the emphasis is $1.7 V_{pp}$. This driving signal was applied to the modulator with a GSG probe. The optical eye diagrams with a data source of PRBS-31 is shown in Fig. 13 and an open eye is observed at 40 Gbps. Although forward bias leads to an increased loss which manifests as a reduced optical ER, it is still possible to obtain a higher modulated ER compared to reverse bias operation because of the extremely low $V_{\pi}L_{\pi}$ of forward bias. The ER measured using the DCA with the optical sampling module was comparable to, or slightly larger than, the previous modulation cases.

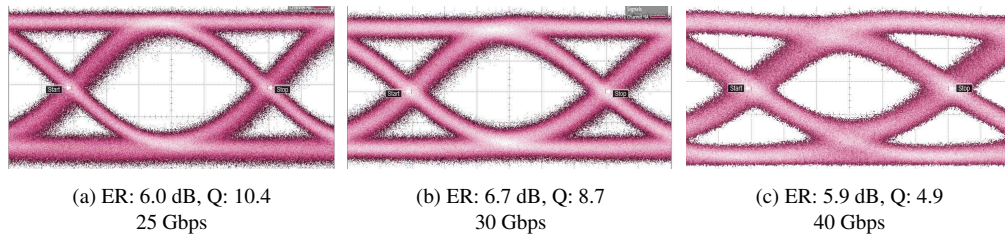


Fig. 13. Optical eye diagrams at different bitrates for the p-i-n modulator operated in forward bias with pre-emphasized PRBS-31 signal.

The measured EO S_{21} response in forward bias at 0.6 V is shown in Fig. 12(c), and it had a 3-dB bandwidth of approximately 0.5 GHz. The calculated response of the FFE with $h[0] = 1$,

$h[1] = -0.944$, and $\tau = 15\text{ps}$, and the calculated total response normalized to the maximum are also shown in Fig. 12(c). With this configuration, the transitions are emphasized by attenuating low frequency content. The estimated low-frequency 3-dB cut-off is 2.3 GHz and the high-frequency 3-dB cut-off is 23.5 GHz at the higher end. Bit error rate testing was performed at 30 Gbps and the results shown in Fig. 12(d) are pattern dependent. Error-free operation was observed for PRBS-15 patterns at -6 dBm, and at -8 dBm for PRBS-7 patterns. Error-free operation was not observed for PRBS-31 signals up to a received power of 3 dBm. This is due to the poor bandwidth at the lower end of the frequency range, since the NRZ spectral content at lower frequencies increase with pattern length.

3.6. Energy estimate

The estimated energy consumption for the three modulation cases demonstrated is shown in Table 2. Under reverse bias operation and neglecting the parasitic elements, energy is consumed by the static leakage current and the dynamic charging and discharging of the junction capacitance ($\Delta E = c_j V^2$). With NRZ signaling and assuming equiprobable bit sequences, there is only one charge-discharge cycle every four bits thus reducing the dynamic energy consumption to $\Delta E = (1/4)c_j V^2$, where V is the driving voltage [24]. We use the extracted capacitance value at the bias voltage for the calculation. The static power consumption for this case is negligible because the current drawn is in the nano-amperes range as indicated in Fig. 3(a). It should be noted that because the modulator does not represent a 50 Ω load, the actual modulating voltage is different than the one measured with a 50 Ω instrument and more accurate energy consumption can be obtained with time-domain reflectometry measurements [25]. In our estimate, the worst possible case of the modulating voltage being two times the driving voltage was assumed. The estimated energy consumed for this driving configuration is listed in column 1 of Table 2.

The power consumption calculation for the modulation case with a parallel resistor is similar to traveling-wave modulators. In this situation, the power is mostly consumed by the resistor, and the power consumption by the modulating junction is small. As with dual-drive traveling-wave modulators, the static power consumption across the termination resistor from reverse bias can be significant. Therefore, it is desirable to operate modulators at lower reverse bias voltages resulting in a bandwidth-power trade-off. The static power was determined by Ohm's law, $P = V_{bias}^2/50\ \Omega$. The dynamic power was calculated assuming equiprobable 1s and 0s and a rectangular driving voltage leading to the expression for the RMS power of a 50% duty cycle square wave: $P = (V_{pp}/2)^2/50\ \Omega$. This power is multiplied by a bit period for energy. The estimated energy consumed for this driving configuration is listed in column 2 of Table 2.

Table 2. Estimated energy consumption of the MIM with reverse biased p-n, reverse biased p-n with a 50 Ω parallel resistor, and forward biased p-i-n with pre-emphasis.

	p-n	p-n (w/ 50 Ω)	Pre-emphasis p-i-n
Dynamic Energy (30 Gbps)	1.92 pJ/bit	1.93 pJ/bit	1.99 pJ/bit
Static Power	5.9 nW	405 mW	17.0 mW
Total Energy (30 Gbps)	1.92 pJ/bit	15.43 pJ/bit	2.56 pJ/bit

The power consumed by pre-emphasis signals is highly pattern dependent, e.g., data with more bit transitions will have a significantly higher power consumption. We estimated the power consumption for the forward bias MIM by capturing a long pattern of the PRBS signal with a real-time oscilloscope, and then computing the RMS voltage. The energy was then calculated as ($\Delta E = c_j V_{rms}^2$). The extracted capacitance at 0.6 V forward bias was determined to be 435.4 fF. Here, we also operated the modulator without a 50 Ω load, thus twice the driving

voltage was used in the calculation. The estimated energy consumed for this driving configuration is listed in column 3 of Table 2.

In this sub-section, the total energy consumption of the same modulator in three different driving configurations was estimated. This novel comparison suggests that the energy consumed by operating the modulator in forward bias is not significantly larger than with reverse bias operation. In addition, it should be noted that the dynamic energy consumption for the cases without a termination can be approximately halved with a dual-drive modulator and that the total energy per bit for all cases will also drop with higher bitrates.

4. Conclusion

We report the DC, small signal, and large signal characterization of a 500- μm lumped Michelson interferometric modulator fabricated in a CMOS process. The p-n junction modulator had a relatively low $V_\pi L_\pi$ of 0.72 V-cm at 1 V reverse bias. Using a driving voltage of $3.4 V_{pp}$, error-free operation was demonstrated up to 40 Gbps in reverse bias. Initially, at 25 Gbps, a received optical power of -5 dBm was required. The net source impedance was then lowered with the use of a 50 Ω RF termination which resulted in a 10 GHz bandwidth increase at 4 V reverse bias. Error-free operation of 25 Gbps with a received optical power of -9.8 dBm was observed, and at 40 Gbps with 0.6 dBm of power. We determined that further reduction in source impedance and the associated transmission line can lead to a significant increase in bandwidth. In forward bias, we measured a very low $V_\pi L_\pi$ of 0.058 V-cm for a p-i-n junction modulator and observed clear eye diagrams up to 40 Gbps using pre-emphasized signals. The modulated ER and the estimated energy consumed by the modulator with forward bias operation is comparable to reverse bias operation. We observed that the performance of such systems depend on the PRBS pattern length indicating the importance of optimizing the emphasized (or equalized) frequency response and the proper choice of test data. Error-free operation in this driving configuration was not observable for PRBS-31 patterns, but was verified with PRBS-7 patterns at a received power of -8 dBm. Several trade-offs associated with energy consumption, bandwidth, and sensitivity were highlighted to outline the vast number of possibilities in testing a silicon photonic communication link.

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